2010 F=ma Exam: Problem 4

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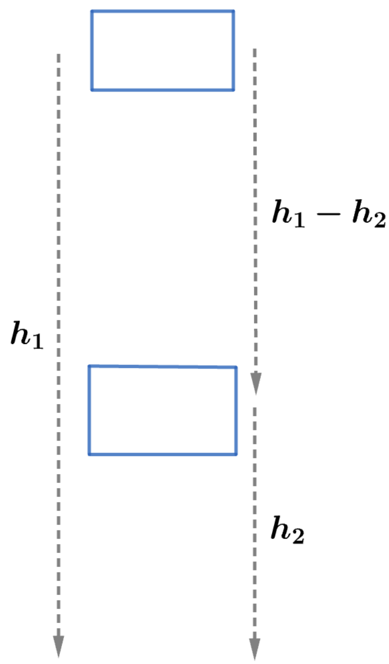

The time taken for the piano to drop from rest at h_1 to the ground is

$$h_1 = \frac{1}{2} g t_{\text{tot}}^2$$
$$t_{\text{tot}} = \sqrt{\frac{2h_1}{g}}$$

The time taken for the piano to drop from rest at h_1 to h_2 is

$$h_1 - h_2 = \frac{1}{2} g t_1^2$$
$$t_1 = \sqrt{\frac{2(h_1 - h_2)}{g}}$$

Thus, the time taken for the piano (released at h_1) to drop from h_2 to the ground is

$$t_2 = t_{\text{tot}} - t_1 = \sqrt{\frac{2h_1}{g}} - \sqrt{\frac{2(h_1 - h_2)}{g}} = \sqrt{\frac{2(30\,\text{m})}{10\,\text{m/s}^2}} - \sqrt{\frac{2(30\,\text{m} - 14\,\text{m})}{10\,\text{m/s}^2}} = 0.66\,\text{s}$$

so the answer is $\boxed{\text{A}}$.